

What is a Preprocessor Directive in C++?

A **preprocessor directive** is a special instruction in a C++ program that is processed **before the actual compilation of the program**.

- It starts with the **# symbol**.
- It does **not** end with a semicolon (;).
- It tells the **preprocessor** to perform a specific task before the compiler compiles the code.

Example

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World";
    return 0;
}
```

Here:

```
#include <iostream>
```

is a **preprocessor directive**.

It tells the preprocessor to include the contents of the **iostream** header file before compilation.

Common Preprocessor Directives

Directive	Purpose
<code>#include</code>	Includes a header file
<code>#define</code>	Defines a constant or macro
<code>#if</code>	Starts a conditional section
<code>#ifdef</code>	Checks whether a macro is defined
<code>#ifndef</code>	Checks whether a macro is not defined

Simple definition for students:

A preprocessor directive is an instruction beginning with **#** that is processed before the C++ program is compiled.